

Cost-Aware Customer Churn Prediction for Telecom Industry

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A. Motivation

Customer churn -- when a paying subscriber leaves a service -- is one of the most expensive problems in telecom. Acquiring a new customer costs 5x-7x more than retaining one, so identifying customers about to leave is directly tied to revenue. Churn is rarely a surprise overnight: usage slips, support tickets spike, bills become unpredictable -- weak signals appear weeks or months before cancel. Carriers already hold the data to spot those patterns; the gap is reliable risk scores and thresholds aligned with who you can afford to save.

For a 1M-subscriber carrier with 25% churn, a 3% reduction = ~7,500 retained customers/year, worth millions in lifetime value.

Why standard ML misses the mark

Imbalanced labels

Only 26.6% of customers churn. A trivial 'predict no-churn' model gets 73% accuracy and zero business value. Accuracy is the wrong metric.

Asymmetric cost

Missing a churner (\$1,500 lost LTV) is ~30x worse than a false alarm (\$50 retention call). Optimization must reflect this.

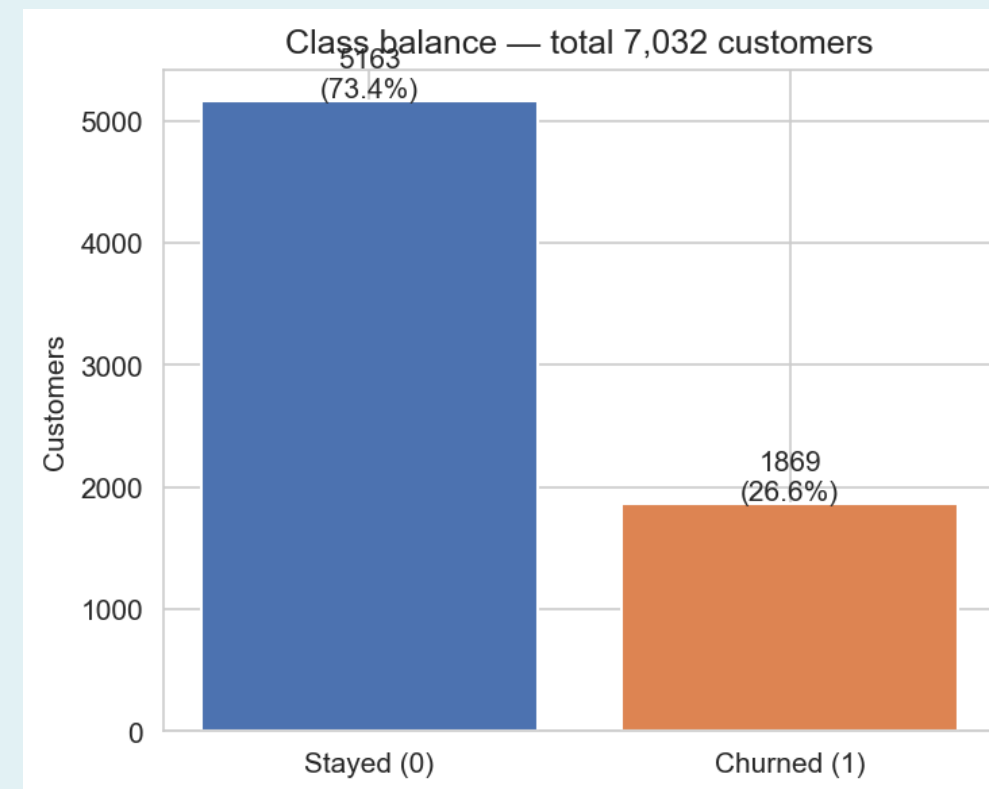
Threshold blindness

Default 0.5 cutoffs ignore both imbalance and cost. A 'good' model with the wrong threshold is still a poor business tool.

B. Data & EDA

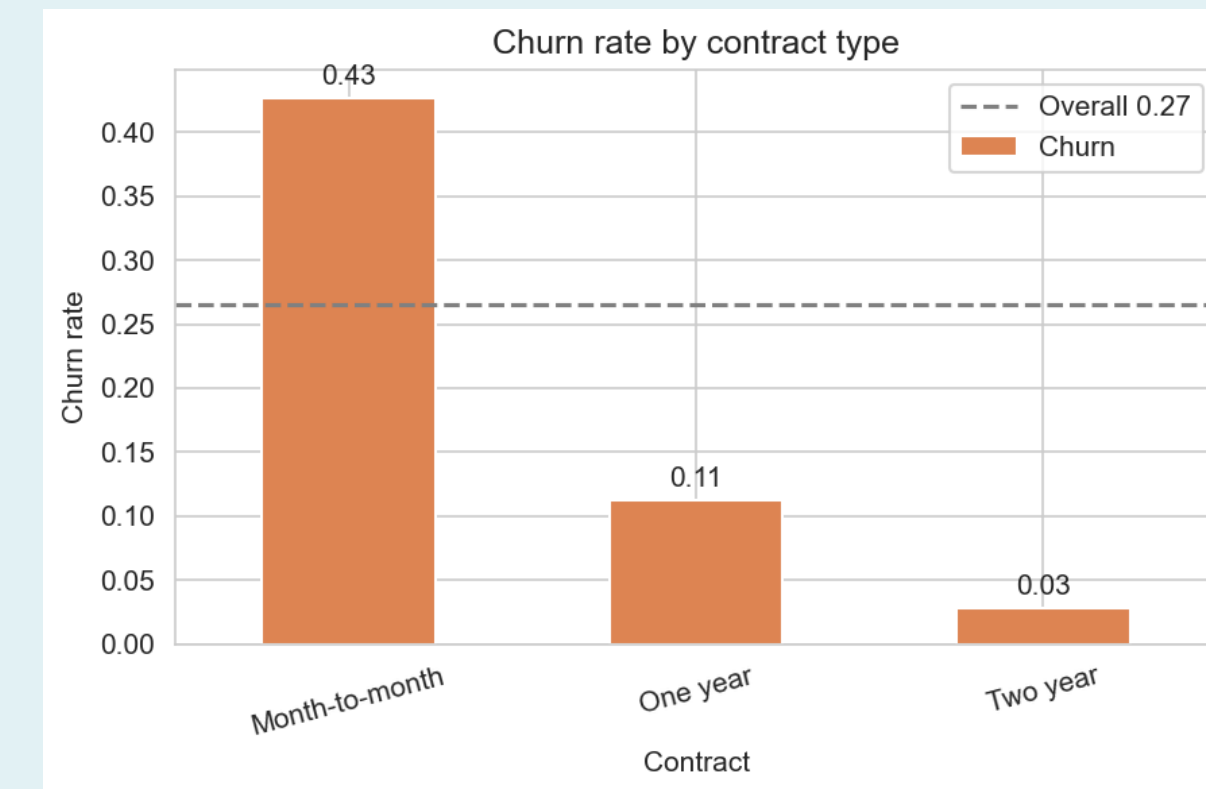
Dataset: IBM Telco Customer Churn (Kaggle)

7,032 customers x 19 features after cleaning (11 dropped for invalid TotalCharges).
Demographics, tenure, contract, services (phone/internet/security/streaming), monthly + total charges. Target: Churn (Yes/No).



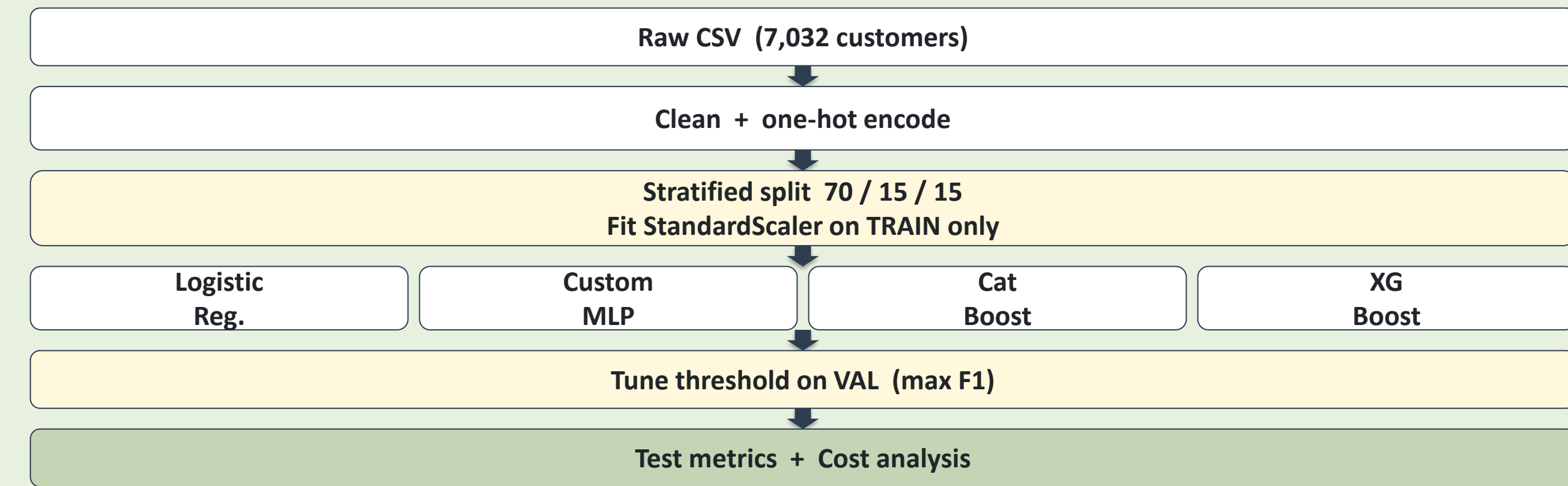
What the data tells us

Imbalanced class: 5,163 stayed (73.4%) vs 1,869 churned (26.6%) -- ratio 1 : 2.76.
Contract dominates: month-to-month 43% vs 2-year 3% -- a 14x gap.
Tenure: churners cluster heavily in the first 12 months.
Service: fiber 42% vs DSL 19%; electronic-check 45% vs auto-pay <20%.



C. Methodology

One disciplined pipeline -- so differences come from architecture, not preprocessing.



Why each step

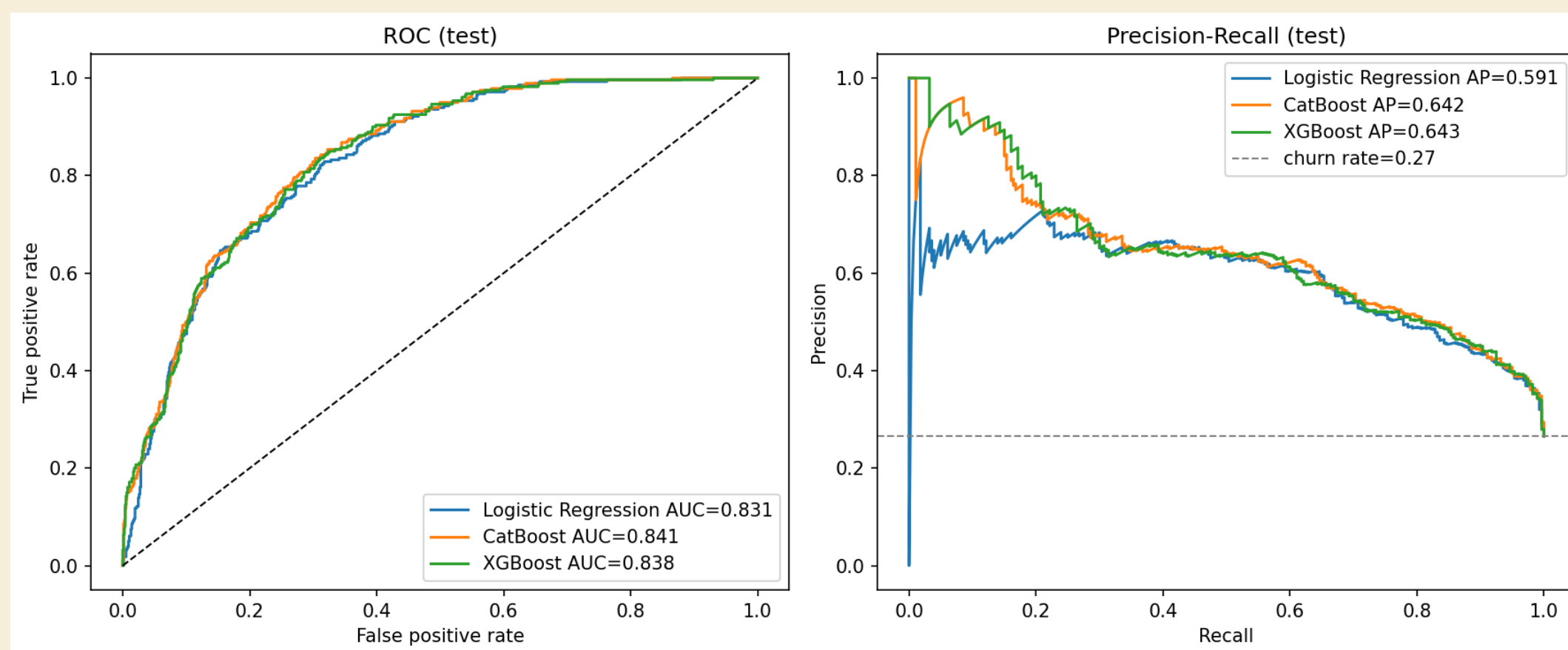
Stratified split: preserves the 26.6% churn rate in every fold.
Train-only scaling: prevents test-set leakage into training.
Threshold tuning: default 0.5 is arbitrary on imbalanced data.
Brier + PR-AUC: measure calibration and minority-class ranking.

D. Results -- Model Comparison

All four models trained on identical features and splits, scored on test with a validation-tuned threshold (max F1).
Brier captures calibration: do predicted probabilities match observed churn rates? Lower is better.

Model	F1	ROC-AUC	PR-AUC	Brier	Recall	Precision
CatBoost	0.624	0.841	0.642	0.139	0.768	0.526
Logistic Regression	0.610	0.831	0.591	0.176	0.746	0.516
XGBoost	0.608	0.838	0.643	0.169	0.725	0.523
MLP (custom NN)	0.603	0.826	0.592	0.177	0.789	0.488

CatBoost wins on F1, ROC-AUC, and Brier -- it ranks well AND its probabilities are calibrated.
MLP does not beat logistic regression -- on small one-hot tabular data, a linear boundary is most of the signal.



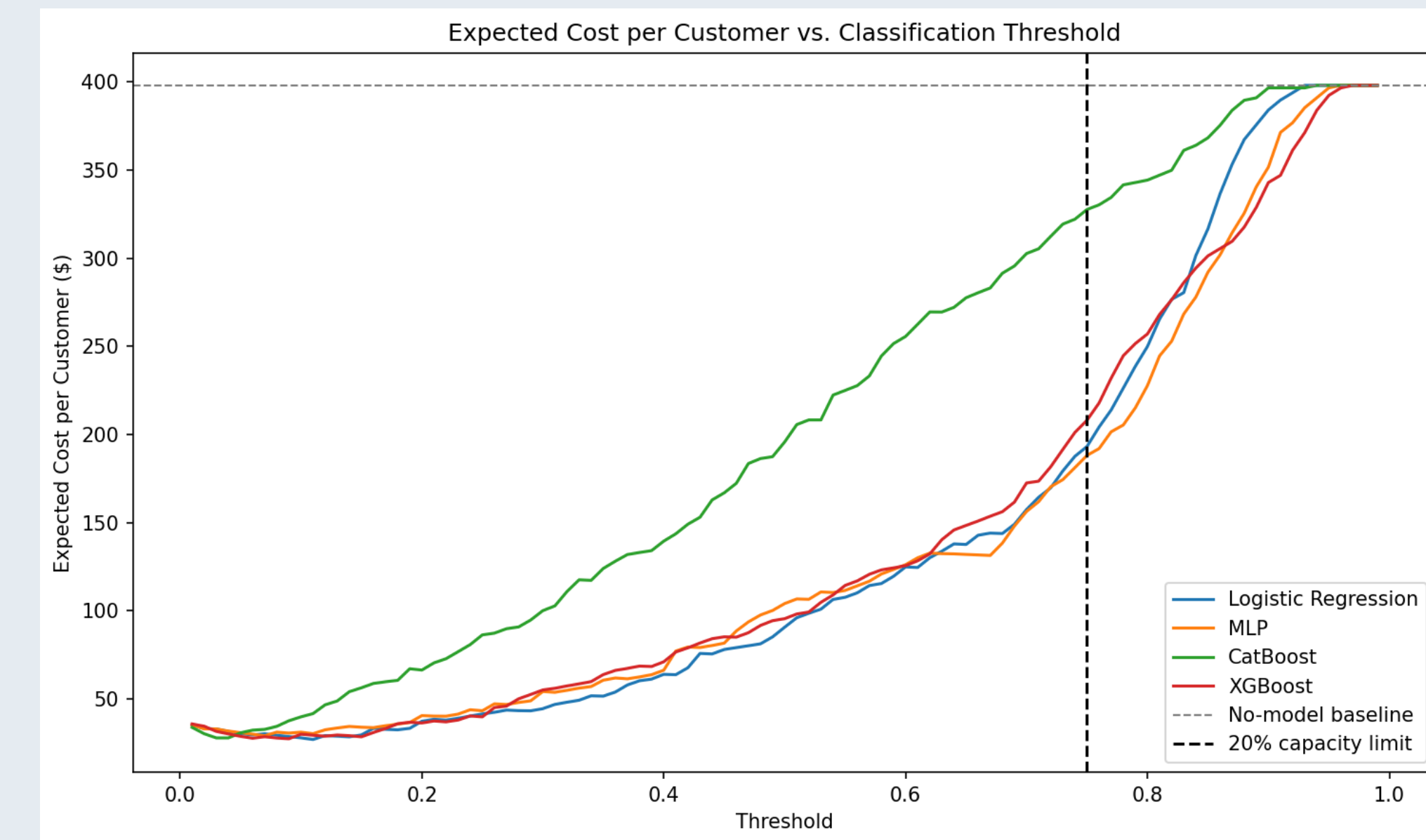
ROC and Precision-Recall curves on test set.

E. Cost-Aware Analysis

Sweep threshold under realistic costs: FN \$1,500 FP \$50. Net Value = TP*\$1500 - FP*\$50. Optimum chosen on validation.

Model	Threshold	Net value / 1,000
Logistic Regression	0.11	\$371,043
XGBoost	0.09	\$370,616
CatBoost	0.03	\$370,142
MLP (custom NN)	0.07	\$368,910

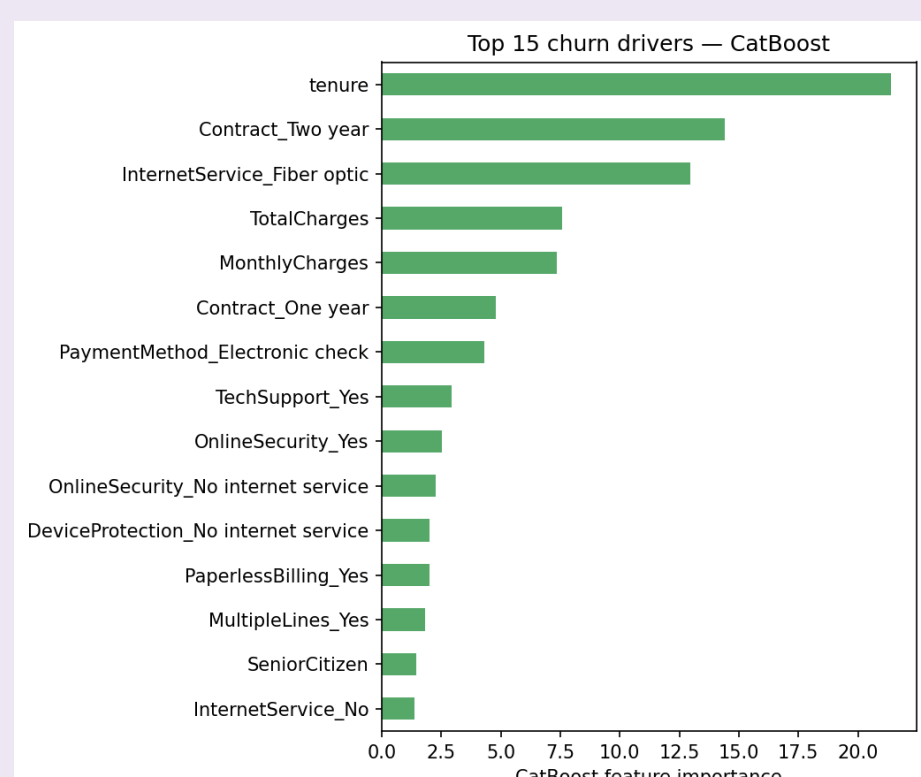
Top-20% capacity constraint: CatBoost still delivers \$175,687 / 1,000 at threshold 0.54.



~ \$370K
net business value
per 1,000
customers when
threshold is tuned

COUNTERFACTUAL	ACTION	RESULT
Customer: predicted churn = 0.93	Switch InternetService: Fiber -> No internet	predicted churn = 0.82 (-11 pp)

F. What Drives Churn



Top driver	Importance
tenure	21.4
Contract = Two year	14.4
InternetService = Fiber optic	13.0
TotalCharges	7.6
MonthlyCharges	7.4
Contract = One year	4.8
PaymentMethod = Electronic check	4.3

Aligns with industry intuition: short tenure, no long contract,
fiber, electronic check -- known retention friction points.
Lever: contract upsell campaigns target the top 3 drivers.

G. Conclusions & Next Steps

Findings

- CatBoost is the strongest classifier (F1 0.624, ROC-AUC 0.841, lowest Brier 0.139).
- Custom MLP does not beat logistic regression -- nonlinearity adds little on small one-hot data.
- Cost-aware threshold tuning is the dominant lever -- all four models recover ~\$370K per 1,000 customers; architecture changes the result by < 1%.
- Counterfactuals turn probabilities into specific retention actions.

Deployment readiness

Save: model + scaler + threshold (0.31).
Score monthly; recalibrate quarterly.

Limitations

Static snapshot; no temporal drift.
Cost ratios illustrative -- vary by segment.

Next steps

Stacking ensemble.
Cost-sensitive loss in MLP.
Live A/B test of threshold.

References & Acknowledgements

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- [3] Chen, T. & Guestrin, C. (2016). XGBoost: a scalable tree-boosting system. KDD.
Dataset: IBM Telco Customer Churn (Kaggle: blastchar/telco-customer-churn). Acknowledgements: Dr. Edgar Lobaton (instructor, ECE 525, NC State University).